Airbyte, Fivetran & DBT Task

October 14 , 2026

Context & Assignment Instructions

Role: Junior Data Engineer

Company: *QuickCart*, a rapidly growing e-commerce startup.

Task: The CTO has tasked you with designing their initial data ingestion strategy. You must write a one-page structural recommendation choosing between **Fivetran** or **Airbyte Open Source (OSS)** based on their specific corporate infrastructure constraints.

Goals

1. Write a one-page structural recommendation choosing between **Fivetran** or **Airbyte Open Source (OSS)** based on their specific corporate infrastructure constraints.

Specifications

Data Specifications & Infrastructure Constraints:

- **Source 1 (High Volume):** A core production PostgreSQL application database. It logs **80 million row updates per month** due to continuous inventory changes, but the data analytics team only builds reports once a day.
- **Source 2 (Long-Tail Custom):** An in-house, custom-built internal inventory API created by a software engineer who left the company last year. No public documentation exists for this API.
- **The Team:** 1 Senior Data Engineer (who handles the entire company's infrastructure infrastructure) and you (the Junior Engineer).
- **The Budget:** Tight on immediate software expenditures, but the company already has plenty of unused AWS EC2 and Kubernetes cloud hosting credits.